Inversion Probability

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1728 | Mathematics | https://cses.fi/problemset/task/1728

An array has n integers $x_1,x_2,\ldots,x_n$, and each of them has been randomly chosen between 1 and r_i. An inversion is a pair (a,b) where $a<b$ and $x_a>x_b$.

What is the expected number of inversions in the array?

Input

The first input line contains an integer n: the size of the array.

The second line contains n integers $r_1,r_2,\ldots,r_n$: the range of possible values for each array position.

Output

Print the expected number of inversions rounded to six decimal places (rounding half to even).

Constraints

- $1 \le n \le 100$
- $1 \le r_i \le 100$

Example

Input
3 5 2 7

Output
1.057143